Soham Rohit Chitnis

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Education

New York University, Courant Institute — **New York, NY, USA**
Master of Science in Computer Science (*GPA: **3.96/4.0***) — September 2024 - May 2026
- Courses: Deep Learning, Natural Language Processing (NLP), DevOps, Computer Vision

Birla Institute of Technology and Science Pilani (BITS Pilani) — **Goa, India**
Bachelor of Engineering in Computer Science, Minor in Data Science — November 2020 – July 2024
- Courses: Machine Learning, Reinforcement Learning, Database Systems, Data Structures & Algorithms

Experience

CILVR Lab, New York University — **New York, NY, USA**
Graduate Student Researcher — January 2025 – Present
- **Advisors:** Wancong (Kevin) Zhang, Prof. Yann LeCun
- Training world models (JEPA & Hierarchical JEPA) using self-supervised learning for robotic control and planning.

APP Center for AI Research, BITS Pilani — **Goa, GA, India**
Undergraduate Student Researcher — February 2022 – May 2024
- **Advisors:** Prof. Ashwin Srinivasan, Prof. Tanmay Verlekar
- Proposed a domain-specific model for feature extraction in the Whole Slide Image classification that achieved state-of-the-art results (96.85% AUC); improved diagnostic confidence, and was published & selected for an oral presentation at IEEE EMBC.
- Developed AutoRef (with TCS Research), an autonomous peer-review multi-agent system where interacting agents generate and critique reviews based on ICLR standards using LangChain and OpenAI API, achieving a 217% improvement in review quality.

Tata Consultancy Services Research — **Pune, MH, India**
Research Intern — September 2023 – December 2023
- **Advisor:** Dr. Manasi Patwardhan
- Engineered an efficient, plug-and-play multimodal model to visually ground a frozen OPT-125M LLM for chart understanding, using only a trainable linear adapter, fine-tuned CLIP with Charts, and benchmarked vision encoders (CLIP, DePlot) on ChartQA.

VC Lab, Ontario Tech University — **Oshawa, ON, Canada**
MITACS Globalink Research Intern — June 2023 – August 2023
- **Advisor:** Prof. Faisal Qureshi
- Proposed a Spatial-Attention CNN Latent Dirichlet VAE for Hyperspectral Unmixing during the fully-funded MITACS Globalink Research Internship, reducing estimation standard deviation by 95% and earning an oral presentation at IEEE IGARSS 2024.

Skills

Languages: Python, C++, C, Java, JavaScript, Bash, SQL **Web & API Development:** REST APIs, FastAPI, HTML/CSS
Machine Learning: PyTorch, TensorFlow, JAX, Scikit-Learn, Pandas, NumPy, HuggingFace, LangChain, DSPy, OpenCV, MLFlow
Developer Tools & DevOps: GIT, Docker, CI/CD, AWS, Linux, Slurm, Prefect
Software & Scientific Computing: MATLAB, LaTeX, MS Office Suite (Excel, Word, PowerPoint)

Selected Projects

Real-time Credit Card Fraud Detection — Code
- Architected an end-to-end AWS **MLOps** pipeline using MLflow and Prefect for model orchestration (73.5% F1-score), while automating CI/CD through Github Actions, FastAPI for real-time inference, and EvidentlyAI for monitoring.

Multi-Task fine-tuning for Machine Translation of Code-Mixed Languages — Poster
- Trained Llama 3.2 1B with QLoRA using Multi-Task Fine-tuning to translate Hindi-English sentences to English and Hindi. Improved BLEU scores by 7.82% & 5.32% for English and Romanized Hindi, respectively, in comparison with SFT.

Action-aware Representation Alignment (AC-REPA) for Navigation World Models — Code Report
- Trained Action Conditioned Diffusion Transformer (CDiT-S) with AC-REPA loss that improves rollout dynamics by 3.9%, semantic similarity at long horizon by 4.5%, and yields closer temporal smoothness to groundtruth by 6.2% with reduced variance by 32%.

Publications

- `AutoRef`: Generating Refinements of Reviews Given Guidelines
  Chitnis S., Patwardhan, M., Srinivasan, A., Verlekar, T.T., Vig, L., Shroff, G. — SDP Workshop, ACL 2024
- SpACNN-LDVAE: Spatial Attention Convolutional Latent Dirichlet Variational Autoencoder for Hyperspectral Pixel Unmixing
  Chitnis S., Mantripragada, K., Qureshi, F.Z. — **(Oral)** IEEE IGARSS 2024
- Domain-Specific Pre-training Improves Confidence in Whole Slide Image Classification
  Chitnis S.R., Liu, S., Dash, T., Verlekar, T.T., Di Ieva, A., Berkovsky, S., Vig, L., Srinivasan, A. — **(Oral)** IEEE EMBC 2023